

Beyond SFT-to-RL: Pre-alignment via Black-Box On-Policy Distillation for Multimodal RL

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Research Questions

- How can we repair the distributional drift introduced by SFT in multimodal models before entering RL optimization?
- Can we design a black-box adversarial alignment method that addresses heterogeneous perception and reasoning errors separately?
- Does an explicit distribution alignment stage between SFT and RLVR consistently improve downstream RL performance across multiple algorithms?

Core Idea

Standard SFT→RL training introduces distributional drift where models neither preserve original capabilities nor match supervision distribution, particularly problematic for multimodal reasoning where perception and reasoning errors compound differently. PRISM inserts an alignment stage using adversarial on-policy distillation with a specialized MoE discriminator that has separate perception and reasoning experts, allowing disentangled correction of visual grounding vs logical reasoning errors without requiring teacher logits.

Key Formula

MoE Discriminator Score

$$r(x, y) = \alpha \cdot D_v(x, c) + (1 - \alpha) \cdot D_r(x, t)$$

Combines weighted scores from perception expert D_v evaluating visual description c and reasoning expert D_r evaluating reasoning trace t , where α controls trade-off between perceptual and reasoning feedback

Adversarial OPD Objective

$$\min_{\theta} \max_{\phi} \mathbb{E}_{(x, y^+) \sim T, y^- \sim G_{\theta}(\cdot|x)} [r_{\phi}(x, y^+) - r_{\phi}(x, y^-)]$$

Minimax game between policy G_{θ} and discriminator with parameters ϕ , where policy minimizes difference between supervision y^+ and policy rollout y^- rewards while discriminator maximizes this gap

Key Figure

Preprint.

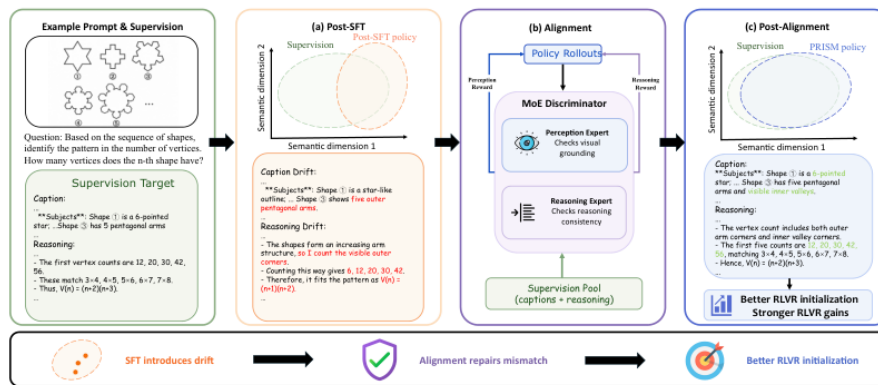


Figure 1: Overview of the PRISM pipeline. (a) SFT introduces distributional drift between the policy and the supervision distribution. (b) The alignment stage uses an MoE discriminator with dedicated perception and reasoning experts to repair this drift via adversarial on-policy distillation. (c) The resulting distribution-aligned policy provides a stronger initialization for downstream RLVR.

Figure 1: Shows complete PRISM pipeline demonstrating how SFT introduces drift, alignment stage repairs mismatch via MoE discriminator, and results in better RLVR initialization with concrete example of perception/reasoning corrections

Limitations

- Additional training cost due to alignment stage and MoE discriminator requiring joint maintenance of policy and discriminator
- Relies on structured response formats (explicit visual caption and reasoning trace) limiting applicability to tasks without natural decomposition
- Distribution alignment analysis uses structural proxies rather than direct distributional measures like embedding-space divergences

- Current MoE design requires predetermined decomposition into perception vs reasoning rather than learned specialization